

Taigo Sakai

Ph.D. Student & Special Assistant, Meijo University
Computer Vision · Continual Learning · Multi-View Detection & Tracking

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RESEARCH PROFILE

I develop learning systems that remain reliable under distribution shift, class imbalance, and limited sensing. My work spans **continual and long-tailed learning**, dynamic neural capacity, and **multi-view pedestrian detection and tracking**, with additional work on robust medical image classification. I value controlled experiments, strong baselines, and methods that improve accuracy without hiding computational or data costs.

EDUCATION & APPOINTMENTS

Ph.D. Student, Meijo University Graduate School of Science and Technology; advisor: Prof. Kazuhiro Hotta.	Apr. 2026 - Present
Special Assistant, Meijo University	Apr. 2026 - Present
M.E., Electrical Engineering, Meijo University	Apr. 2024 - Mar. 2026
B.E., Electrical and Electronic Engineering, Meijo University	Apr. 2020 - Mar. 2024

PEER-REVIEWED INTERNATIONAL PUBLICATIONS

- 2026 **T. Sakai**, K. Hotta, “**NeuroGuard: Neural Gradient Update Aware of Representation Damage.**” ECCV 2026 Workshop on Human-inspired Computer Vision. **Oral.**
- 2026 K. Toida, **T. Sakai**, T. Nakamura, H. Shimizu, K. Hotta, “**ATLASFusion: Aggregation Tracking with Location-Aware Sparse Fusion for Robust Spatio-Temporal Multi-View Pedestrian Tracking.**” ECCV 2026 Workshop on Computer Vision for Multimedia Spatial Intelligence through Time. Poster.
- 2026 **T. Sakai**, K. Hotta, “**Dynamic Distillation and Gradient Consistency for Robust Long-Tailed Incremental Learning.**” ICIP 2026. Poster.
- 2026 **T. Sakai**, K. Hotta, “Improving Noise Robustness in Medical Image Classification via Dynamic Integration of CNN and KAN.” CARS 2026. Poster.
- 2026 **T. Sakai**, T. Nakamura, H. Shimizu, K. Hotta, “**Visibility-Gated ConvGRU for Robust Multi-View Pedestrian Detection and Tracking in BEV Space.**” VISAPP 2026. **Oral.**
- 2025 **T. Sakai**, K. Hotta, “**Long-Tailed Data Classification by Increasing and Decreasing Neurons During Training.**” ICCV 2025 Workshop on Human-inspired Computer Vision. Poster.
- 2024 **T. Sakai**, K. Hotta, “**Growing Deep Neural Network Considering with Similarity between Neurons.**” ECCV 2024 Workshop on Human-inspired Computer Vision. Poster.

SELECTED HONORS

- 2026 DIA2026 Research Encouragement Award - 3 selected from 96 papers.
- 2026 IEICE Tokai Branch Student Research Encouragement Award.
- 2026 Rinnai Scholarship Foundation - selected scholarship recipient.
- 2025 ViEW2025 Visual Inspection Algorithm Contest - **First Prize** (1st of 131 entries).
- 2025 MIRU2025 Student Encouragement Award (55 selected from 726 submissions).
- 2024 IPSJ CVIM Best Bachelor’s Thesis Session Award.
- 2021-25 Meijo University President’s Award - received in 2021, 2023, 2024, and 2025.

ADDITIONAL RESEARCH PRESENTATIONS

2026 MIRU 2026 - Gabor-KAN for medical image classification. Peer-reviewed poster.
2026 DIA 2026 - Visibility-Gated ConvGRU for multi-view pedestrian detection and tracking. Peer-reviewed oral.
2026 FIT 2026 - Rational-function-basis KAN for MedViTV2 medical image classification. Oral.
2025 MIRU 2025 - Gradient consistency regularization and dynamic knowledge distillation for long-tailed continual learning. Peer-reviewed oral.
2025 DIA 2025 - Plant rhizoid tip tracking from limited supervision. Peer-reviewed poster.
2025 SSII 2025 - Multi-view multi-object tracking. Peer-reviewed poster.
2024 ViEW 2024 - Increasing/decreasing neurons during training for imbalanced classification. Peer-reviewed poster.
2024 IPSJ CVIM 238 - Bachelor thesis session presentation on neuron growth with similarity distributions.

RESEARCH & PROFESSIONAL EXPERIENCE

Special Assistant, Meijo University Research and academic support in computer vision and deep learning.	2026 - Present
R&D; Intern, SECOM Co., Ltd. Development Internships Fuso Koki; HiKOKI.	Aug. - Sep. 2024 2022

ACADEMIC SERVICE

Technical Program Committee - ECCV 2026 3rd Human-inspired Computer Vision Workshop.
Technical Program Committee - ICCV 2025 2nd Human-inspired Computer Vision Workshop.
Technical Program Committee - ECCV 2024 Human-inspired Computer Vision Workshop.
MIRU 2026 Wakate Program - survey poster and presentation on new-task research.
Nagoya CV/PRML Study Group - research paper presentations (2024).

SELECTED ENGINEERING WORK

iOS/Web application development - shipped and maintained multiple applications spanning AI-assisted fashion, presentation preparation, image super-resolution, NFC photo albums, and productivity tools.
Selected projects and source code: [portfolio](#) · github.com/sakai1250

CERTIFICATIONS

JDLA Deep Learning Engineer, Japan Deep Learning Association - Aug. 2024.
Third-Class Electric Chief Engineer - Mar. 2023.

RESEARCH KEYWORDS

Continual Learning · Long-Tailed Recognition · Dynamic Neural Capacity · Knowledge Distillation · Multi-View Detection · Multi-Object Tracking · BEV Representation · Robust Learning · Medical Image Classification